Two-Board Architecture for Hardware Fault Injection

We adopted a two-board setup for fault injection, using the Nexys A7-100T as the injector and the VC707 as the target. This physical separation allows us to conduct real hardware-level fault injection experiments that would be difficult, if not impossible, to reproduce through software emulation or on a single-board platform. Below, we explain why this design offers a more practical and versatile fault injection environment.

1. Physical Separation and Cleaner Fault Channels

Why it matters:
Keeping the attacker and target on separate FPGAs gives us a clean and controllable fault injection path. The A7 connects to the VC707 through PMOD lines, which we use to drive faults into clock or input lines.
For example:

- Stuck-at faults are easy to generate by simply holding a PMOD line high or low,
- Instruction corruption can be emulated by toggling bus lines during fetch cycles,
- Timing violations emerge when we send unsynchronized pulses that interfere with the VC707's internal clock edges,
- We can implement bit-flip attacks as fast glitches or pulse trains that collide with sampling windows.

Attempting this on a single board would bring complications. We'd have to work around internal clock routing, avoid interfering with the global clock tree, and consume more FPGA resources for glitch generation. Using two boards keeps the VC707 design uninfected.

2. Independent Clock Domains

Real-world relevance:
The A7 and VC707 each run on their respective oscillators (typically 100 MHz, but configurable). This independence lets us simulate real external interference. Since the A7's pulses are unsynchronized with the VC707's clock, we can trigger edge-case behavior, like metastability or setup/hold violations, that wouldn't surface in a shared-clock setup.
More importantly, we can test how faults behave statistically across varying phase offsets or mismatched frequencies (e.g., running the A7 at 75 MHz or 95 MHz). This behavior is valuable for exploring fault timing sensitivity in actual asynchronous systems.

3. Modular and Reprogrammable Injection Logic

Scalability through decoupling:
Keeping the injector logic on the A7 lets us build reusable modules for different fault types:

- A stuck-at module that holds signals high or low,
- An instruction-glitch unit that selectively flips instruction bits mid-fetch,
- A timing glitch generator that produces adjustable pulse trains,
- A bit-flip injector that delivers fast glitches to I/O or bus lines.

Since all fault logic stays on the A7, we can freely reprogram it without touching the VC707. The target remains consistent, typically running a RISC-V softcore under Linux or bare-metal code, while the A7 acts as a flexible test driver.

4. Safety and Debug Flexibility

Protecting hardware and workflows:
A separate injector board adds a layer of protection. If something goes wrong, for example, a glitch shorts a line or locks up the logic, it only affects the A7, not the VC707. This separation reduces the risk of damaging expensive or irreplaceable hardware.
On the software side, we can run simulations (e.g., in QEMU) alongside physical experiments. If QEMU predicts that a stuck-at-1 fault on register 'x10' causes a kernel panic, we can replicate the same fault on the VC707 and observe the result. This parallelism helps us cross-validate results between emulation and hardware.

5. Trade-offs and Practical Considerations

What this approach costs, and why it's worth it:

There are obvious overheads: more hardware, more wiring, more pin coordination, and a steeper integration curve. Additionally, we need to carefully manage the voltage levels and pin assignments on both platforms.

But in return, we get:

- Cleaner signal control,
- Independent clock domains,
- Greater modularity,
- And real physical effects, like rise time, propagation delay, and I/O signal distortion, that emulators can't replicate.

For serious fault injection studies, this trade is more than fair.

Conclusion

Using the Nexys A7 as a dedicated fault injector and the VC707 as the fault target gives us a powerful and flexible hardware testbed. This two-board architecture enables precise and varied fault injections without compromising the target design or realism. It's a practical and scalable setup that captures physical effects software models alone cannot match.